

ASSESSMENT 3:

Subject Title & Code	Web Design and Programming (BIS111)
Lecturer	Rashikala Weerawarna
Assessment Title	Report and Presentation (Group)
Assessment Due Date	Week 13- 16 June 2026 11:59 PM
Assessment Weighting	35%
Assessment Submission	Brightspace (group submission)
Assessment File Name	A3_BIS111_Assessment3_YourName
Other Assessment Requirements	
1. This is a group assessment.	
2. Use an appropriate academic writing style and referencing format as specified. Use Arial, 11-point font and 1.5 line spacing. Include page numbers. Reference list on a separate page.	
3. If you use the ideas, models or words of others they must be properly referenced. (plagiarism is a serious offence and can result in harsh penalties)	
4. In this assessment, you are PERMITTED to use generative artificial intelligence, written declaration must be submitted with the assessment. (Attached as an appendix). All use of AI must be referenced using Holmesglen APA 7th.	
5. Submission should be through the Learning Management System (Brightspace) and should consist of a single ZIP file.	
6. Late submission 5% per day penalty on the final mark of the assessment.	
7. By submitting the assessment via Brightspace, you are confirming that the work is your own (and your group's) and that a backup copy has been retained.	
Assessment Instructions	
Learning Outcomes Assessed	
This assessment is mapped to the following learning outcomes for Web Design and Programming (BIS111):	
• LO3: Construct a responsive web application • LO4: Design web pages and integrate with database structures • LO5: Identify the advantages and disadvantages of using a web development framework	
Assessment Tasks:	
You are required to take one group members assignment 2 and work on this.	
You are to improve upon this and work as a team	
- Improve on the UI - Improve on the functionality - Each team member needs to contribute 1 new CRUD view with required functionality - The new view can be anything, as long as it is part of the domain i.e. part of a shopping cart or store. - Store all team work in a GitHub repository - All team members should have access to the repository and need to submit the work. - All team members will need to present the group project at the end of the project - Each member will need to talk for at least 5 minutes.	

This structured approach will guide you and your group through the complete process of developing the selected management system, ensuring you focus on all the necessary aspects of software development.

Task 1: SDLC Planning:

Identify and outline the phases of the Software Development Life Cycle (SDLC) you will follow for this project. Provide a brief explanation of each phase and describe how it applies to the development of your management system.

Task 2: Project Requirements

Based on the chosen domain (Gym and Fitness, Real Estate, Coffee Shop, Pet Care, or Student Management), list and describe the functional and non-functional requirements of your system. What are the key features that your system must include? How will these requirements meet the needs of the domain?

Task 3: Pseudo Code

Develop pseudo code for the main functionalities of your system. This should include CRUD operations for the core entities and any specific logic needed to manage associations between entities (e.g., linking clients to properties or students to courses).

Task 4: Flow Chart

Create a flow chart that visually represents the workflow of your system. The flow chart should depict the process for a user interacting with the system, from input to processing and output. Include key decision points and processes for CRUD operations and data validation.

Task 5: Database Normalization

Normalize your database design to at least the third normal form (3NF). Provide a step-by-step explanation of how you arrived at the final database structure. Ensure that your design eliminates redundancy and enforces data integrity.

Task 6: Entity-Relationship Diagram (ERD)

Design an Entity-Relationship Diagram (ERD) for your system. The ERD should clearly show the entities, their attributes, and the relationships between them, including primary and foreign keys. Explain how the ERD supports the requirements of your system.

Task 7: Coding

Implement the system based on your designs and pseudo code. The implementation should include:

- CRUD operations for all entities.
- Data management in a connected mode using ADO.NET.
- Appropriate exception handling.
- A user interface (if applicable) to interact with the system.
- Version control using GitHub, with regular commits documenting your progress.

Task 8: Testing and Debugging

Write and execute both system and unit tests for your application. The tests should cover all critical functionalities, including CRUD operations, entity associations, and error handling. Document any bugs found during testing and describe how you resolved them.

Task 9: Publishing

Deploy your application to a suitable environment. This could be a local server, cloud platform, or any other hosting environment. Provide a detailed guide on how to set up and run your application in the chosen environment.

Task 10: Documentation

Prepare comprehensive documentation for your project. This should include:

- System architecture and design.
- Database schema and ERD.
- User guide explaining how to use the system.
- Instructions for setting up and running the system.
- Summary of the testing approach and results.

Task 11: Presentation

Present your completed system to an audience. Your presentation should cover:

- An overview of the project and its purpose.
- Key features and how they address the domain's needs.
- The design process, including SDLC, pseudo code, flow chart, and ERD.
- A demonstration of the system in action.
- A summary of testing and debugging efforts.
- Reflection on what you learned and any challenges you encountered.

This assignment carries 100% and will be converted to 35% marks of the overall mark for the subject. The marking for this assignment will be done as follows:

Rubric for Assessment Tasks (Team of 3 Members, if more or less than 3 members please adjust workloads fairly)

Task	Weight (%)	Assigned Member	Criteria
Task 1: SDLC Planning	10%	Team Leader	Criteria: Clarity of SDLC phases, relevance to the project, and application of SDLC to project tasks.
Task 2: Project Requirements	5%	Member(s) name	Criteria: Completeness of functional and non-functional requirements, alignment with project objectives, and understanding of domain needs.
Task 3: Pseudo Code	5%	Member(s) name	Criteria: Accuracy and detail in pseudo code, logical structure, and coverage of main functionalities (CRUD, entity associations).
Task 4: Flow Chart	10%	Member(s) name	Criteria: Correctness and clarity of the flow chart, logical representation of workflow, and inclusion of key processes and decision points.
Task 5: Database Normalisation	5%	Member(s) name	Criteria: Correct application of normalization principles (3NF), clear explanation of the process, and effectiveness in reducing redundancy.
Task 6: ERD	10%	Member(s) name	Criteria: Accuracy of the ERD, proper representation of relationships between entities, and inclusion of primary and foreign keys.
Task 7: Coding	20%	Member(s) name	Criteria: Correctness and functionality of the code, adherence to design and pseudo code, use of connected mode, and quality of exception handling.

Task 8: Testing and Debugging	10%	Member(s) name	Criteria: Thoroughness of testing approach, variety and relevance of test cases, effectiveness in debugging, and clarity in documenting bugs and fixes.
Task 9: Deployment	5%	Member(s) name	Criteria: Completeness of deployment steps, correctness of setup instructions, and functionality of the deployed system.
Task 10: Documentation	10%	All members	Criteria: Completeness of system documentation, clarity of user guide, and comprehensiveness of testing summary.
Task 11: Presentation	10%	All members	Criteria: Effectiveness of the presentation, coverage of key project aspects, clarity of communication, and team member involvement.

